Industrial IoT Energy Management SaaS

Master Project Blueprint (Version 1)

Prepared as a handoff document for future AI models and developers.

1. Project Goal

Build a production-grade, multi-tenant Industrial IoT SaaS platform starting with textile processing factories. The system must monitor machines, ingest telemetry, generate analytics, support predictive maintenance, and later include OCR, costing and AI modules.

2. Business Context

Initial deployment targets textile dyeing and finishing. Target assets include Jet Dyeing Machines, Stenters, Boilers, Thermic Fluid Heater, Pumps, Fans, Blowers, Compressors and Utilities.

3. Technical Stack

Backend: FastAPI
ORM: SQLAlchemy
Database: PostgreSQL
Deployment: Docker
Future: Redis, MQTT, Grafana, Flutter, Firebase Notifications.

4. Multi-tenant Design

Every company is isolated. Core hierarchy:
Company → Departments → Machines → Machine Components → Telemetry Tags → Telemetry Data.

5. Current Completed Work

✓ Dockerized backend
✓ FastAPI deployed on DigitalOcean
✓ Swagger documentation (/docs)
✓ Authentication foundation
✓ Company, User, Department, Machine, Machine Component models
✓ Initial telemetry model
✓ GitHub repository initialized
✓ Project documentation started.

6. Database Philosophy

Reference/master tables define reusable templates. Instance tables represent actual factory equipment. Telemetry is stored separately to support high-volume time-series ingestion.

7. Planned Modules

• Device Provisioning
• MQTT Collector
• Modbus Polling Engine
• Alarm Engine
• Notification Engine
• Dashboard Builder
• Predictive Maintenance
• Energy Reports
• Production Reports
• Water Monitoring
• Recipe OCR
• Costing Engine

8. API Standards

REST endpoints with validation. JWT authentication. Versioned APIs. Backward compatibility. Swagger generated from FastAPI.

9. Coding Standards

Always add comments. Preserve existing APIs. Never delete working code without approval. Document major changes. Explain code for learning. Keep production-ready quality.

10. AI Instructions

Any AI working on this project should first read the documentation, preserve the architecture, avoid breaking schema compatibility, and update documentation after each major feature.

11. Future Roadmap

Phase 1: Stable backend
Phase 2: MQTT ingestion
Phase 3: Telemetry pipeline
Phase 4: Dashboards
Phase 5: Flutter apps
Phase 6: Predictive maintenance
Phase 7: Textile OCR and costing
Phase 8: SaaS scaling to thousands of factories.

12. Long-Term Vision

The platform should become a generic Industrial IoT operating system that supports multiple industries while keeping textile modules as optional domain packages.